

Explainable AI Dashboard Components Guide

Project Overview (Context)

Title

Explainable AI System for Research Gap Detection and Emerging Topic Analysis

Core problem addressed

Researchers struggle to manually review rapidly growing literature, identify meaningful gaps, track emerging topics, and query their own corpus with grounded evidence.

Core idea used by this system

A research gap is modeled as two semantically similar topics with low co-occurrence, then explained with measurable evidence (similarity, co-occurrence, and supporting papers).

Why this dashboard exists

The dashboard is the operational UI layer that turns embedding, clustering, gap detection, trend analysis, map projection, and RAG QA into an explainable end-to-end workflow.

Dashboard Shell

1) Dashboard Page Container

- File: `src/pages/dashboard/page.tsx`
- Consists of:
 - Sidebar navigation
 - Top bar
 - Main section renderer (Overview, Datasets, Gap Detection, Topic Explorer, Trend Analysis, Research Map, AI Chatbot)
 - Results full-screen overlay
 - Analysis history panel
- Why keep it:
 - Acts as orchestration layer for the whole dashboard.
 - Centralizes section routing/state and preserves a consistent UX frame.
 - Connects analysis runs to history and results without page reload.

2) Sidebar

- File: `src/pages/dashboard/components/Sidebar.tsx`
- Consists of:
 - Workspace navigation entries
 - Records area (Analysis History trigger + badge)
 - User identity and logout

- Why keep it:
 - Gives stable task navigation across all analysis modules.
 - Keeps primary workflow (datasets -> gaps/topics/trends/map/chat) one click away.
 - Preserves session awareness with analysis history count.

3) Top Bar

- File: `src/pages/dashboard/components/TopBar.tsx`
 - Consists of:
 - Contextual section title/subtitle
 - Search input
 - Quick action buttons: snapshots, analysis history, export history
 - Related panels (ExportHistoryPanel, SnapshotsPanel, AnalysisHistoryPanel)
 - Why keep it:
 - Keeps global actions available independent of selected section.
 - Improves explainability and reproducibility via snapshots/history.
-

Sidebar Pages: What They Contain and Why They Are Kept

1) Overview

- Main file: `src/pages/dashboard/sections/OverviewSection.tsx`
- Consists of:
 - Key stats cards (papers, topics, gaps, average gap score)
 - Latest run summary banner
 - Top research gaps preview
 - Recent papers panel
 - Topic distribution cards with trend indicators
- Why keep it:
 - Provides fast situational awareness before deep analysis.
 - Supports your objective of reducing manual effort by surfacing high-value findings first.

2) Datasets

- Main file: `src/pages/dashboard/sections/DatasetsSection.tsx`
- Consists of:
 - Paper management list and filtering/sorting
 - Processing trigger for analysis pipeline
 - Pipeline modal integration
 - Export helpers (JSON/CSV/print)
 - Run metadata creation for downstream results/history
- Why keep it:
 - This is the intake and execution gateway for the entire AI workflow.
 - Connects user-uploaded corpus to explainable analysis outputs.

3) Gap Detection

- Main file: `src/pages/dashboard/sections/GapsSection.tsx`

- Consists of:
 - Gap cards ranked by score
 - Filter controls (similarity, gap score, sorting)
 - Select mode for compare/export
 - Gap detail side panel and compare modal
 - Export (JSON/PDF)
- Why keep it:
 - This is the core value component tied directly to your problem statement.
 - Makes gap reasoning transparent with score, similarity, co-occurrence, and evidence.

4) Topic Explorer

- Main file: `src/pages/dashboard/sections/TopicsSection.tsx`
- Consists of:
 - Topic exploration UI
 - Topic detail modal with keywords and linked papers
 - Topic compare modal (overlap, shared papers, trend alignment)
- Why keep it:
 - Supports explainability by exposing cluster semantics and paper-level grounding.
 - Helps verify that gap candidates are meaningful, not random noise.

5) Trend Analysis

- Main file: `src/pages/dashboard/sections/TrendsSection.tsx`
- Consists of:
 - Trend cards with mini time-series charts
 - Growth labels (rising/stable/declining)
 - Timeline annotations (first-study and bridge events)
 - Trend filtering by status
- Why keep it:
 - Directly addresses your objective for emerging topic detection.
 - Adds temporal evidence so decisions are not based on static snapshots.

6) Research Map

- Main file: `src/pages/dashboard/sections/MapSection.tsx`
- Consists of:
 - 2D topic/paper map (UMAP-like projection view)
 - Centroid and gap-line visualization controls
 - Citation view mode and citation edge highlighting
 - Export options (DOT/JSON graph outputs)
- Why keep it:
 - Translates complex relationships into interpretable spatial structure.
 - Supports explainability and discovery of sparse/under-explored regions.

7) AI Chatbot

- Main file: `src/pages/dashboard/sections/ChatbotSection.tsx`

- Consists of:
 - Session list and active chat panel
 - Prompt input and quick suggestions
 - Answer generation with citation blocks and relevance bars
 - Loading states and message formatting
 - Why keep it:
 - Delivers natural language interaction over the corpus.
 - Aligns with your RAG objective: grounded responses with evidence.
-

Supporting Analysis Pages (Not Current Sidebar Entries)

Results Overlay

- Main file: `src/pages/dashboard/sections/ResultsSection.tsx`
- Consists of:
 - Unified output sections: dataset, topics, gaps, trends, map, chat, evaluation summary
 - Methodology panel (step-by-step explainability)
 - Save snapshot modal and snapshots panel integration
 - Share/print/export style utilities
- Why keep it:
 - Consolidates pipeline outputs into one evidence report view.
 - Critical for presenting explainable end-to-end analysis in academic settings.

Evaluation Section

- Main file: `src/pages/dashboard/sections/EvaluationSection.tsx`
 - Consists of:
 - Circular metric cards (coherence, coverage, novelty, model quality)
 - Dataset and gap detection performance stats
 - Co-occurrence matrix visualization
 - Why keep it:
 - Provides transparent model quality indicators.
 - Supports defendability of results using quantitative diagnostics.
-

Key Reusable Components in Dashboard Flow

Processing Pipeline Modal

- File: `src/pages/dashboard/sections/datasets/ProcessingPipeline.tsx`
- Role:
 - Simulates staged processing (upload -> extract -> embed -> cluster -> detect gaps)
 - Provides progress visibility and completion callback to results workflow
- Why keep it:
 - Makes analysis lifecycle observable and understandable for users.

Analysis History Panel

- File: `src/pages/dashboard/components/AnalysisHistoryPanel.tsx`
- Role:
 - Lists saved runs and lets users reopen past results
- Why keep it:
 - Supports reproducibility and iterative research workflow.

Snapshots and Export History Panels

- Files:
 - `src/pages/dashboard/components/SnapshotsPanel.tsx`
 - `src/pages/dashboard/components/ExportHistoryPanel.tsx`
 - Role:
 - Track checkpoints and generated outputs
 - Why keep them:
 - Strengthen traceability, reporting, and explainable auditability.
-

Data and Hooks Layer (Why It Matters)

Hooks

- `src/hooks/useAnalysisHistory.tsx`
- `src/hooks/useSnapshots.tsx`
- `src/hooks/useExportHistory.tsx`
- `src/hooks/useAuth.tsx`

These keep dashboard state persistent and structured (runs, snapshots, exports, user auth), which is essential for practical explainable analysis workflows.

Mock Data Sources (current implementation)

- `src/mocks/papers.ts`
- `src/mocks/topics.ts`
- `src/mocks/gaps.ts`
- `src/mocks/trends.ts`
- `src/mocks/evaluation.ts`
- `src/mocks/mapData.ts`
- `src/mocks/chatData.ts`
- `src/mocks/citations.ts`

These files provide deterministic sample outputs for each analysis layer and make UI behavior testable before full backend integration.

Summary

Your dashboard is structured correctly for your proposed system:

- Datasets section handles corpus intake and pipeline kickoff.
- Gap Detection is the core explainable value module.

- Topic, Trend, and Map sections provide multi-layer evidence.
- Chatbot provides grounded interaction over corpus knowledge.
- Results, Evaluation, History, Snapshots, and Export support transparency and academic defendability.